Inequalities – 1

The Cauchy–Bunyakovsky inequality as a technique

Homework

10 сентября 2026 года

Theoretical minimum. The discrete, integral and inner product forms and the proof through the discriminant of $\int (f-\lambda g)^2$; the equality case as proportionality; the Engel form $\sum a_i^2/b_i \geqslant (\sum a_i)^2/\sum b_i$; the choice of the splitting $h = fg$ dictated by the equality case; the Gram determinant.

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1. (RMO 1997) Prove that $\int_{-1}^{1} f^2 \geqslant \frac{1}{2}\big(\int_{-1}^{1} f\big)^2 + \frac{3}{2}\big(\int_{-1}^{1} xf\big)^2$ for every continuous f, and identify the functions for which equality holds.

2. (Putnam 2004) Let $f(x,y)$ be continuous on the unit square. Show that $\int_0^1 \big(\int_0^1 f(x,y)\,dx\big)^2 dy + \int_0^1 \big(\int_0^1 f(x,y)\,dy\big)^2 dx \leqslant \big(\int_0^1\int_0^1 f\big)^2 + \int_0^1\int_0^1 f^2$.

3. (Putnam 2013) For continuous f on $[0,1]$ put $\mu(f) = \int_0^1 f$, $\operatorname{Var}(f) = \int_0^1 (f-\mu(f))^2$ and $M(f) = \max_{[0,1]} |f|$. Show that $\operatorname{Var}(fg) \leqslant 2\operatorname{Var}(f)M(g)^2 + 2\operatorname{Var}(g)M(f)^2$.

4. (RMO 2008) Let $f \in C^1[0,1]$ with $f\big(\frac{1}{2}\big) = 0$. Prove that $\int_0^1 (f')^2 \geqslant 12\big(\int_0^1 f\big)^2$.

5. (RMO 2025) Let f be differentiable on $[0,1]$ with integrable derivative and $f(1) = 0$. Prove that $\int_0^1 (xf'(x))^2 dx \geqslant 12\big(\int_0^1 xf(x)\,dx\big)^2$, and compare the weight with the previous problem.

6. (IMC) Let (x_i) be a decreasing sequence of positive numbers.

   a) Show that $\big(\sum_{i=1}^n x_i^2\big)^{1/2} \leqslant \sum_{i=1}^n \frac{x_i}{\sqrt{i}}$.

   b) Show that there is a constant C with $\sum_{m=1}^{\infty} \frac{1}{\sqrt{m}}\big(\sum_{i=m}^{\infty} x_i^2\big)^{1/2} \leqslant C\sum_{i=1}^{\infty} x_i$.

7. (Петрополитана 2024) Prove that $\int_0^1 e^f \cdot \int_0^1 e^{-f} \geqslant 1 + \int_0^1 f^2 - \left(\int_0^1 f\right)^2$ for every continuous $f\colon [0,1] \to \mathbb{R}$.

8. (ФКН, весна 2023) Let $f\colon \mathbb{R} \to \mathbb{R}_+$ be continuous with period T. Prove that $\int_0^T \frac{f(x)}{f(x+\alpha)}\,dx \geqslant T$ for every real α.

9. Prove that the Gram determinant of any finite family of vectors in an inner product space is non-negative, deduce the Cauchy–Bunyakovsky inequality from the case of two vectors, and interpret the general case as a statement about volumes.

10. (SEEMOUS 2018) Let $f\colon [0,1] \to (0,1)$ be Riemann integrable. Show that
$$\frac{2\int_0^1 x f^2(x)\,dx}{\int_0^1 \left(f^2(x)+1\right)dx} < \frac{\int_0^1 f^2(x)\,dx}{\int_0^1 f(x)\,dx}.$$